Final paper results Sept 24

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Data overview

Our data set includes 4259 unique public water systems representing all of the active CWS and NTNCWS water systems in the same. Of these water systems 2811 are community water systems, 1448 are Non-Transient Non-Community systems. The mean population served by these systems is 9557.4444705.

Describing carceral water systems

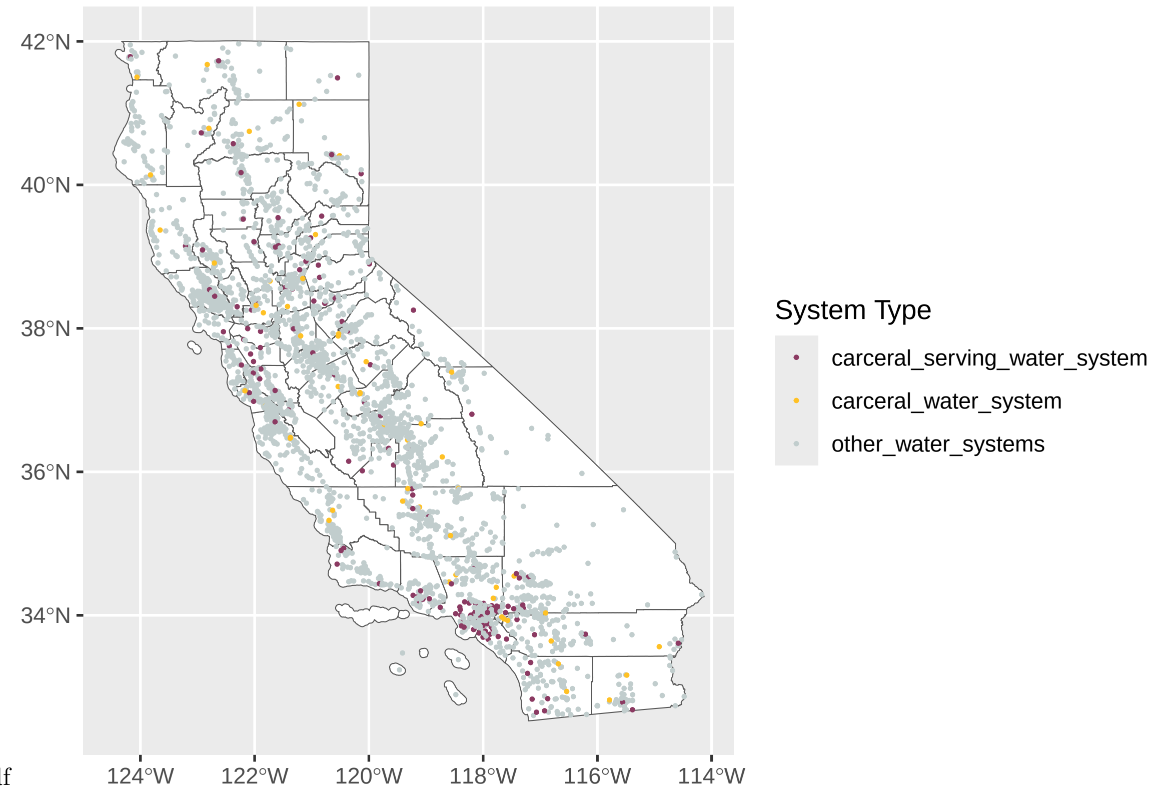
We classify these systems into three categories: Carceral water systems that primarily or exclusively serve carceral facilities; non-exclusive carceral water systems that serve one or more carceral facilities in addition to other customers and non-carceral systems that do not serve any carceral faciltiies. We find that 56 or 1.3148626% are carceral water systems. 151 or 3.5454332% are non-exclusive carceral systems and 4052 or 95.1397042% are noncarceral systems.

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Tue, Oct 07, 2025 - 10:13:03

Table 1:

Statistic	N	Mean	St. Dev.	Min	Max

As you can see above, many of the carceral serving systems rely on surface water. Which is interesting. Many of those systems are concentrated in the bay area, Sacramento and northern central valley areas and southern california whereas exclusive carceral systems are fairly evenly spread throughout the state.



map-1.pdf

Outcome (performance) descriptive results

Next we can see if these categories of water systems perform differently using different performance measures related to water quality, accessibility and TMF capacity.

Chi squared and GLMs for outcomes

Lastly we can use GLMs and chi squared tests to see if any of these relationships are statistically significant while holding other driving factors like system size constant.

```
MCL <- xtabs(~ health_5yr_no2ndary_or_lsl_binary + carceral_sys , data = Data)
chisq.test(MCL, correct = FALSE) #no sign dif between any groups
```

```
##
##  Pearson's Chi-squared test
##
## data:  MCL
## X-squared = 3.1889, df = 2, p-value = 0.203
```

```
MON <- xtabs(~ mon_only_5yr_binary + carceral_sys , data = Data); MON
```

```
##                    carceral_sys
## mon_only_5yr_binary carceral_serving_water_system carceral_water_system
##                   0                           116                    32
##                   1                            35                    24
##                    carceral_sys
## mon_only_5yr_binary other_water_systems
##                   0                2735
##                   1                1317
```

```r
chisq.test(MON, correct = FALSE) #sig p=0.01163
```

```
##
##  Pearson's Chi-squared test
##
## data:  MON
## X-squared = 8.6739, df = 2, p-value = 0.01308
```

```r
chisq.test(MON[,c(2,1)]) #carceral serving and carceral sig different p=0.008
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  MON[, c(2, 1)]
## X-squared = 6.8268, df = 1, p-value = 0.00898
```

```r
chisq.test(MON[,c(2,3)]) # carceral and other not sig (p=0.1342)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  MON[, c(2, 3)]
## X-squared = 2.2432, df = 1, p-value = 0.1342
```

```r
chisq.test(MON[,c(1,3)]) #carceral serving and other sig different p=0.01793
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  MON[, c(1, 3)]
## X-squared = 5.3803, df = 1, p-value = 0.02037
```

```r
CCR <- xtabs(~ Missing_ccr_any + carceral_sys , data = Data); CCR
```

```
##                carceral_sys
## Missing_ccr_any carceral_serving_water_system carceral_water_system
##             No                            110                    35
##             Yes                            41                    21
##                carceral_sys
## Missing_ccr_any other_water_systems
##             No                 2208
##             Yes                1844
```

```r
chisq.test(CCR, correct = FALSE) # Sig differences p<0.001
```

```
##
##  Pearson's Chi-squared test
##
## data:  CCR
## X-squared = 21.046, df = 2, p-value = 2.691e-05
```

```r
chisq.test(CCR[,c(2,1)]) # Carceral serving and carceral not different
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  CCR[, c(2, 1)]
## X-squared = 1.6208, df = 1, p-value = 0.203
```

```r
chisq.test(CCR[,c(2,3)]) # Carceral and other systems not different
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  CCR[, c(2, 3)]
## X-squared = 1.1243, df = 1, p-value = 0.289
```

```r
chisq.test(CCR[,c(1,3)]) #carceral serving and other systems sig dif p <0.001
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  CCR[, c(1, 3)]
## X-squared = 19.096, df = 1, p-value = 1.243e-05
```

```r
Data_notassessedremoved <- Data
Data_notassessedremoved$Water_Quality_Risk_Level_Display[Data$Water_Quality_Risk_Level_Display == "Not A
Data_notassessedremoved$Accessability_Risk_Level_Display[Data$Accessability_Risk_Level_Display == "Not A
Data_notassessedremoved$TMF_Capacity_Risk_Level_Display[Data$TMF_Capacity_Risk_Level_Display == "Not Ass
Data_notassessedremoved$SAFER_STATUS[Data$SAFER_STATUS == "Not Assessed"] <- NA
Outcomes_notassessedremoved <- Data_notassessedremoved[,c(8,9,11,23,24,26,28,37)]
Outcomes_notassessedremoved <- na.omit(Outcomes_notassessedremoved)
Outcomes_notassessedremoved <- droplevels(Outcomes_notassessedremoved)

QR <- xtabs(~ Water_Quality_Risk_Level_Display + carceral_sys , data = Outcomes_notassessedremoved); QR
```

```
##                                 carceral_sys
## Water_Quality_Risk_Level_Display carceral_serving_water_system
##              High to medium risk                            28
##              Low to no risk                                 64
##                                 carceral_sys
## Water_Quality_Risk_Level_Display carceral_water_system other_water_systems
##              High to medium risk                    16                 816
##              Low to no risk                         28                2073
```

```r
chisq.test(QR, correct = FALSE) # not sig different p=0.4513
```

```
##
##  Pearson's Chi-squared test
##
## data:  QR
## X-squared = 1.5914, df = 2, p-value = 0.4513
```

```r
chisq.test(QR[,c(2,1)]) # not sig p=0.62
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  QR[, c(2, 1)]
## X-squared = 0.24554, df = 1, p-value = 0.6202
```

```r
chisq.test(QR[,c(2,3)]) # not sig p=0.3091
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
```

```
## data:  QR[, c(2, 3)]
## X-squared = 1.0346, df = 1, p-value = 0.3091
```

```
chisq.test(QR[,c(1,3)]) #not sig p=0.7328
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  QR[, c(1, 3)]
## X-squared = 0.11656, df = 1, p-value = 0.7328
```

```
AR <- xtabs(~ Accessability_Risk_Level_Display + carceral_sys , data = Outcomes_notassessedremoved); AR
```

```
##                                   carceral_sys
## Accessability_Risk_Level_Display carceral_serving_water_system
##              High to medium risk                            15
##              Low to no risk                                 77
##                                   carceral_sys
## Accessability_Risk_Level_Display carceral_water_system other_water_systems
##              High to medium risk                    24                1541
##              Low to no risk                         20                1348
```

```
chisq.test(AR, correct = FALSE) # there are sig differneces, p<0.001
```

```
##
##  Pearson's Chi-squared test
##
## data:  AR
## X-squared = 49.113, df = 2, p-value = 2.164e-11
```

```
chisq.test(AR[,c(2,1)]) #Carceral serving and carceral sig different p<0.001
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  AR[, c(2, 1)]
## X-squared = 19.453, df = 1, p-value = 1.031e-05
```

```
chisq.test(AR[,c(2,3)]) # carceral and other not sig different p=0.99
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  AR[, c(2, 3)]
## X-squared = 4.6237e-05, df = 1, p-value = 0.9946
```

```
chisq.test(AR[,c(1,3)]) #carceral serving and other sig p<0.01
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  AR[, c(1, 3)]
## X-squared = 47.541, df = 1, p-value = 5.387e-12
```

```
TMFR <- xtabs(~ TMF_Capacity_Risk_Level_Display + carceral_sys , data = Outcomes_notassessedremoved); TM
```

```
##                                  carceral_sys
## TMF_Capacity_Risk_Level_Display carceral_serving_water_system
##             High to medium risk                             7
```

```
##             Low to no risk                                 85
##                                   carceral_sys
## TMF_Capacity_Risk_Level_Display carceral_water_system other_water_systems
##             High to medium risk                       33                 384
##             Low to no risk                            11                2505
```

```r
chisq.test(TMFR, correct = FALSE) # there are sig differneces, p<0.001
```

```
##
##  Pearson's Chi-squared test
##
## data:  TMFR
## X-squared = 140.17, df = 2, p-value < 2.2e-16
```

```r
chisq.test(TMFR[,c(2,1)]) #Carceral serving and carceral sig different p<0.001
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  TMFR[, c(2, 1)]
## X-squared = 61.906, df = 1, p-value = 3.603e-15
```

```r
chisq.test(TMFR[,c(2,3)]) # carceral and other are sig different p<0.01
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  TMFR[, c(2, 3)]
## X-squared = 130.3, df = 1, p-value < 2.2e-16
```

```r
chisq.test(TMFR[,c(1,3)]) #carceral serving and other not sig p=0.1519
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  TMFR[, c(1, 3)]
## X-squared = 2.0528, df = 1, p-value = 0.1519
```

```r
STATUS <- xtabs(~ SAFER_STATUS + carceral_sys , data = Outcomes_notassessedremoved); STATUS
```

```
##                         carceral_sys
## SAFER_STATUS             carceral_serving_water_system carceral_water_system
##   At-Risk                                           15                    18
##   Failing                                            6                     2
##   Not At-Risk                                       56                    16
##   Potentially At-Risk                               15                     8
##                         carceral_sys
## SAFER_STATUS             other_water_systems
##   At-Risk                                596
##   Failing                                390
##   Not At-Risk                           1473
##   Potentially At-Risk                    430
```

```r
chisq.test(STATUS, correct = FALSE) # there are sig differneces, p=0.00364
```

```
##
##  Pearson's Chi-squared test
##
```

```
## data:  STATUS
## X-squared = 19.331, df = 6, p-value = 0.00364
```

```r
chisq.test(STATUS[,c(2,1)]) #Carceral serving and carceral sig different p=0.01139
```

```
##
##  Pearson's Chi-squared test
##
## data:  STATUS[, c(2, 1)]
## X-squared = 11.062, df = 3, p-value = 0.01139
```

```r
chisq.test(STATUS[,c(2,3)]) # carceral and other are sig different p=004117
```

```
##
##  Pearson's Chi-squared test
##
## data:  STATUS[, c(2, 3)]
## X-squared = 13.255, df = 3, p-value = 0.004117
```

```r
chisq.test(STATUS[,c(1,3)]) #carceral serving and other not sig different p=0.1166
```

```
##
##  Pearson's Chi-squared test
##
## data:  STATUS[, c(1, 3)]
## X-squared = 5.9002, df = 3, p-value = 0.1166
```

GLMS have controls for population served, water source, if water is purchased and water system type as well as hydrologic region

```r
Data$carceral_sys <- relevel(Data$carceral_sys, ref = "other_water_systems")

MCL_violations <- glm(health_5yr_no2ndary_or_lsl_binary ~ carceral_sys + population_served_count + water
summary(MCL_violations)
```

```
##
## Call:
## glm(formula = health_5yr_no2ndary_or_lsl_binary ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                               -1.669e+00  1.292e-01 -12.918
## carceral_syscarceral_serving_water_system  1.141e-01  3.168e-01   0.360
## carceral_syscarceral_water_system         -3.301e-01  4.170e-01  -0.792
## population_served_count                   -4.661e-06  2.574e-06  -1.811
## watersourceSW                              3.911e-01  1.447e-01   2.702
## purchasedPurchased                        -1.219e+00  2.435e-01  -5.004
## pws_type_codeNTNCWS                       -1.304e-01  1.227e-01  -1.063
## HR_NAMEColorado River                      6.297e-02  2.832e-01   0.222
## HR_NAMENorth Coast                        -5.436e-01  2.177e-01  -2.497
## HR_NAMENorth Lahontan                     -5.802e-01  4.483e-01  -1.294
## HR_NAMESacramento River                   -7.599e-01  2.077e-01  -3.658
## HR_NAMESan Francisco Bay                  -7.963e-01  3.082e-01  -2.583
## HR_NAMESan Joaquin River                   4.766e-01  1.629e-01   2.925
## HR_NAMESouth Coast                        -1.316e-01  2.157e-01  -0.610
```

```
## HR_NAMESouth Lahontan                       1.224e-01  2.202e-01   0.556
## HR_NAMETulare Lake                          8.724e-01  1.623e-01   5.376
##                                            Pr(>|z|)
## (Intercept)                                 < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.718826
## carceral_syscarceral_water_system         0.428602
## population_served_count                   0.070201 .
## watersourceSW                             0.006884 **
## purchasedPurchased                         5.6e-07 ***
## pws_type_codeNTNCWS                       0.287935
## HR_NAMEColorado River                     0.824036
## HR_NAMENorth Coast                        0.012515 *
## HR_NAMENorth Lahontan                     0.195572
## HR_NAMESacramento River                   0.000254 ***
## HR_NAMESan Francisco Bay                  0.009781 **
## HR_NAMESan Joaquin River                  0.003440 **
## HR_NAMESouth Coast                        0.541697
## HR_NAMESouth Lahontan                     0.578299
## HR_NAMETulare Lake                         7.6e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2974.0  on 3441  degrees of freedom
## Residual deviance: 2791.5  on 3426  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 2823.5
##
## Number of Fisher Scoring iterations: 6
```

```
M_violations <- glm(mon_only_5yr_binary ~ carceral_sys + population_served_count + watersource + purcha
summary(M_violations)
```

```
##
## Call:
## glm(formula = mon_only_5yr_binary ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -6.107e-01  9.891e-02  -6.174
## carceral_syscarceral_serving_water_system -1.971e-01  2.319e-01  -0.850
## carceral_syscarceral_water_system          2.815e-01  2.840e-01   0.991
## population_served_count                   -1.931e-06  1.381e-06  -1.398
## watersourceSW                             -3.490e-01  1.173e-01  -2.977
## purchasedPurchased                        -2.832e-01  1.532e-01  -1.848
## pws_type_codeNTNCWS                       -3.064e-01  9.913e-02  -3.090
## HR_NAMEColorado River                      1.030e+00  2.117e-01   4.865
## HR_NAMENorth Coast                         1.404e-01  1.473e-01   0.953
## HR_NAMENorth Lahontan                     -5.228e-01  3.188e-01  -1.640
## HR_NAMESacramento River                    2.068e-02  1.368e-01   0.151
## HR_NAMESan Francisco Bay                  -2.629e-01  1.931e-01  -1.362
## HR_NAMESan Joaquin River                  -1.984e-01  1.362e-01  -1.457
```

```
## HR_NAMESouth Coast                         2.024e-01  1.524e-01   1.328
## HR_NAMESouth Lahontan                      4.300e-01  1.647e-01   2.611
## HR_NAMETulare Lake                        -3.137e-02  1.394e-01  -0.225
##                                          Pr(>|z|)
## (Intercept)                              6.66e-10 ***
## carceral_syscarceral_serving_water_system  0.39532
## carceral_syscarceral_water_system          0.32154
## population_served_count                    0.16207
## watersourceSW                              0.00292 **
## purchasedPurchased                         0.06455 .
## pws_type_codeNTNCWS                        0.00200 **
## HR_NAMEColorado River                    1.15e-06 ***
## HR_NAMENorth Coast                         0.34072
## HR_NAMENorth Lahontan                      0.10106
## HR_NAMESacramento River                    0.87985
## HR_NAMESan Francisco Bay                   0.17324
## HR_NAMESan Joaquin River                   0.14518
## HR_NAMESouth Coast                         0.18405
## HR_NAMESouth Lahontan                      0.00902 **
## HR_NAMETulare Lake                         0.82191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4326.7  on 3441  degrees of freedom
## Residual deviance: 4223.9  on 3426  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 4255.9
##
## Number of Fisher Scoring iterations: 5
```

```r
MR_violations <- glm(mr_5yr_binary ~ carceral_sys + population_served_count + watersource + purchased +
summary(MR_violations)
```

```
##
## Call:
## glm(formula = mr_5yr_binary ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                              -8.624e-02  9.348e-02  -0.923
## carceral_syscarceral_serving_water_system -1.944e-01  2.138e-01  -0.909
## carceral_syscarceral_water_system          2.289e-01  2.796e-01   0.819
## population_served_count                  -2.846e-06  1.374e-06  -2.072
## watersourceSW                            -4.441e-01  1.084e-01  -4.096
## purchasedPurchased                       -2.497e-01  1.404e-01  -1.778
## pws_type_codeNTNCWS                      -2.289e-01  9.100e-02  -2.516
## HR_NAMEColorado River                     1.084e+00  2.203e-01   4.920
## HR_NAMENorth Coast                        1.839e-01  1.398e-01   1.316
## HR_NAMENorth Lahontan                    -6.357e-01  2.901e-01  -2.191
## HR_NAMESacramento River                   3.331e-01  1.290e-01   2.581
## HR_NAMESan Francisco Bay                 -5.327e-02  1.726e-01  -0.309
```

```
## HR_NAMESan Joaquin River                     -1.389e-01  1.264e-01  -1.099
## HR_NAMESouth Coast                            1.371e-01  1.449e-01   0.947
## HR_NAMESouth Lahontan                         2.572e-01  1.602e-01   1.605
## HR_NAMETulare Lake                            1.088e-01  1.306e-01   0.833
##                                              Pr(>|z|)
## (Intercept)                                   0.35624
## carceral_syscarceral_serving_water_system  0.36313
## carceral_syscarceral_water_system             0.41295
## population_served_count                       0.03828 *
## watersourceSW                                4.21e-05 ***
## purchasedPurchased                            0.07542 .
## pws_type_codeNTNCWS                           0.01188 *
## HR_NAMEColorado River                        8.64e-07 ***
## HR_NAMENorth Coast                            0.18827
## HR_NAMENorth Lahontan                         0.02842 *
## HR_NAMESacramento River                       0.00984 **
## HR_NAMESan Francisco Bay                      0.75761
## HR_NAMESan Joaquin River                      0.27160
## HR_NAMESouth Coast                            0.34386
## HR_NAMESouth Lahontan                         0.10846
## HR_NAMETulare Lake                            0.40458
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4745.4  on 3441  degrees of freedom
## Residual deviance: 4617.2  on 3426  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 4649.2
##
## Number of Fisher Scoring iterations: 5
```

```r
Data_notassessedremoved <- droplevels(Data_notassessedremoved)
Data_notassessedremoved_small <- Data_notassessedremoved[, c(8,23,24,26,28,6,17,18,3,19)]
Data_notassessedremoved_small <- Data_notassessedremoved_small %>% drop_na(Water_Quality_Risk_Level_Disp
Data_notassessedremoved_small$Water_Quality_Risk_Level_Display <- relevel(Data_notassessedremoved_small$
Data_notassessedremoved_small$Accessability_Risk_Level_Display <- relevel(Data_notassessedremoved_small$
Data_notassessedremoved_small$TMF_Capacity_Risk_Level_Display <- relevel(Data_notassessedremoved_small$T
Data_notassessedremoved_small$carceral_sys <- relevel(Data_notassessedremoved_small$carceral_sys, ref =
Data_notassessedremoved_small$SAFER_STATUS <- factor(Data_notassessedremoved_small$SAFER_STATUS, ordered

NA_WQ <- glm(Water_Quality_Risk_Level_Display ~ carceral_sys + population_served_count + watersource + p
summary(NA_WQ)
```

```
##
## Call:
## glm(formula = Water_Quality_Risk_Level_Display ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data_notassessedremoved_small)
##
## Coefficients:
##                                                Estimate Std. Error z value
## (Intercept)                                  -6.360e-01  1.064e-01  -5.977
## carceral_syscarceral_serving_water_system  7.825e-02  2.694e-01   0.290
```

```
## carceral_syscarceral_water_system         2.005e-01  3.264e-01   0.614
## population_served_count                   1.100e-05  3.515e-06   3.129
## watersourceSW                            -8.029e-01  1.522e-01  -5.277
## purchasedPurchased                       -3.336e-01  1.900e-01  -1.755
## pws_type_codeNTNCWS                       3.208e-01  1.634e-01   1.963
## HR_NAMEColorado River                     4.801e-01  2.281e-01   2.105
## HR_NAMENorth Coast                       -1.437e+00  2.188e-01  -6.567
## HR_NAMENorth Lahontan                    -1.160e+00  4.005e-01  -2.896
## HR_NAMESacramento River                  -5.683e-01  1.561e-01  -3.640
## HR_NAMESan Francisco Bay                 -1.009e+00  2.721e-01  -3.710
## HR_NAMESan Joaquin River                 -1.178e-01  1.484e-01  -0.794
## HR_NAMESouth Coast                        1.620e-01  1.645e-01   0.985
## HR_NAMESouth Lahontan                     1.030e-01  1.831e-01   0.562
## HR_NAMETulare Lake                        3.436e-01  1.505e-01   2.284
##                                          Pr(>|z|)
## (Intercept)                              2.27e-09 ***
## carceral_syscarceral_serving_water_system 0.771475
## carceral_syscarceral_water_system        0.539125
## population_served_count                  0.001755 **
## watersourceSW                            1.31e-07 ***
## purchasedPurchased                       0.079174 .
## pws_type_codeNTNCWS                      0.049682 *
## HR_NAMEColorado River                    0.035280 *
## HR_NAMENorth Coast                       5.14e-11 ***
## HR_NAMENorth Lahontan                    0.003783 **
## HR_NAMESacramento River                  0.000272 ***
## HR_NAMESan Francisco Bay                 0.000207 ***
## HR_NAMESan Joaquin River                 0.427346
## HR_NAMESouth Coast                       0.324711
## HR_NAMESouth Lahontan                    0.573832
## HR_NAMETulare Lake                       0.022396 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3427.9  on 2868  degrees of freedom
## Residual deviance: 3205.6  on 2853  degrees of freedom
##   (156 observations deleted due to missingness)
## AIC: 3237.6
##
## Number of Fisher Scoring iterations: 4
```

```r
NA_AC <- glm(Accessability_Risk_Level_Display ~ carceral_sys + population_served_count + watersource + p
summary(NA_AC)
```

```
##
## Call:
## glm(formula = Accessability_Risk_Level_Display ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data_notassessedremoved_small)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                               3.030e-02  1.034e-01   0.293
```

```
## carceral_syscarceral_serving_water_system -1.693e-01  3.495e-01  -0.484
## carceral_syscarceral_water_system          2.218e-01  3.262e-01   0.680
## population_served_count                   -9.776e-05  9.860e-06  -9.914
## watersourceSW                              3.687e-01  1.258e-01   2.932
## purchasedPurchased                         1.459e-01  1.692e-01   0.862
## pws_type_codeNTNCWS                        1.738e+00  2.243e-01   7.749
## HR_NAMEColorado River                      2.185e-02  2.333e-01   0.094
## HR_NAMENorth Coast                        -6.598e-02  1.609e-01  -0.410
## HR_NAMENorth Lahontan                     -1.219e+00  3.439e-01  -3.545
## HR_NAMESacramento River                    2.325e-01  1.418e-01   1.639
## HR_NAMESan Francisco Bay                  -2.504e-01  2.129e-01  -1.176
## HR_NAMESan Joaquin River                   5.329e-01  1.462e-01   3.646
## HR_NAMESouth Coast                        -7.150e-01  1.723e-01  -4.148
## HR_NAMESouth Lahontan                     -5.933e-01  1.808e-01  -3.282
## HR_NAMETulare Lake                         1.208e+00  1.673e-01   7.223
##                                           Pr(>|z|)
## (Intercept)                               0.769432
## carceral_syscarceral_serving_water_system 0.628087
## carceral_syscarceral_water_system         0.496384
## population_served_count                    < 2e-16 ***
## watersourceSW                             0.003369 **
## purchasedPurchased                        0.388623
## pws_type_codeNTNCWS                       9.23e-15 ***
## HR_NAMEColorado River                     0.925381
## HR_NAMENorth Coast                        0.681712
## HR_NAMENorth Lahontan                     0.000393 ***
## HR_NAMESacramento River                   0.101154
## HR_NAMESan Francisco Bay                  0.239658
## HR_NAMESan Joaquin River                  0.000267 ***
## HR_NAMESouth Coast                        3.35e-05 ***
## HR_NAMESouth Lahontan                     0.001032 **
## HR_NAMETulare Lake                        5.09e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3977.1  on 2868  degrees of freedom
## Residual deviance: 3340.4  on 2853  degrees of freedom
##   (156 observations deleted due to missingness)
## AIC: 3372.4
##
## Number of Fisher Scoring iterations: 6
```

```r
NA_TMF <- glm(TMF_Capacity_Risk_Level_Display ~ carceral_sys + population_served_count + watersource + p
summary(NA_TMF)
```

```
##
## Call:
## glm(formula = TMF_Capacity_Risk_Level_Display ~ carceral_sys +
##     population_served_count + watersource + purchased + pws_type_code +
##     HR_NAME, family = binomial, data = Data_notassessedremoved_small)
##
## Coefficients:
##                                             Estimate Std. Error z value
```

```
## (Intercept)                               -2.180e+00  1.737e-01 -12.554
## carceral_syscarceral_serving_water_system -1.294e-01  4.290e-01  -0.302
## carceral_syscarceral_water_system          2.771e+00  3.597e-01   7.705
## population_served_count                   -2.347e-05  7.460e-06  -3.146
## watersourceSW                             -6.293e-02  1.676e-01  -0.375
## purchasedPurchased                        -7.958e-01  2.587e-01  -3.076
## pws_type_codeNTNCWS                       -3.008e+00  7.144e-01  -4.211
## HR_NAMEColorado River                      1.487e+00  2.839e-01   5.235
## HR_NAMENorth Coast                         4.092e-01  2.539e-01   1.612
## HR_NAMENorth Lahontan                      1.926e-01  4.820e-01   0.399
## HR_NAMESacramento River                    6.493e-01  2.161e-01   3.005
## HR_NAMESan Francisco Bay                  -1.938e-01  3.948e-01  -0.491
## HR_NAMESan Joaquin River                   7.911e-01  2.166e-01   3.652
## HR_NAMESouth Coast                         9.127e-01  2.382e-01   3.831
## HR_NAMESouth Lahontan                      6.690e-01  2.592e-01   2.581
## HR_NAMETulare Lake                         7.181e-01  2.275e-01   3.157
##                                           Pr(>|z|)
## (Intercept)                                < 2e-16 ***
## carceral_syscarceral_serving_water_system 0.762973
## carceral_syscarceral_water_system         1.31e-14 ***
## population_served_count                   0.001654 **
## watersourceSW                             0.707297
## purchasedPurchased                        0.002098 **
## pws_type_codeNTNCWS                       2.55e-05 ***
## HR_NAMEColorado River                     1.65e-07 ***
## HR_NAMENorth Coast                        0.106934
## HR_NAMENorth Lahontan                     0.689551
## HR_NAMESacramento River                   0.002655 **
## HR_NAMESan Francisco Bay                  0.623593
## HR_NAMESan Joaquin River                  0.000260 ***
## HR_NAMESouth Coast                        0.000128 ***
## HR_NAMESouth Lahontan                     0.009861 **
## HR_NAMETulare Lake                        0.001597 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2392.8  on 2868  degrees of freedom
## Residual deviance: 2175.3  on 2853  degrees of freedom
##   (156 observations deleted due to missingness)
## AIC: 2207.3
##
## Number of Fisher Scoring iterations: 7
```

```r
library(MASS)
Data_notassessedremoved_small_nanone  <- na.omit(Data_notassessedremoved)
SAFER_STATUS <- polr(SAFER_STATUS ~ carceral_sys + population_served_count + watersource + purchased + p
```

```
## Call:
## polr(formula = SAFER_STATUS ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, data = Data_notassessedremoved_small_nanone)
##
## Coefficients:
## carceral_syscarceral_water_system   carceral_sysother_water_systems
```

```
##                  -3.187744e-01                      1.350168e-01
##          population_served_count                    watersourceSW
##                   2.679844e-05                      5.541753e-02
##              purchasedPurchased                pws_type_codeNTNCWS
##                   3.592371e-01                     -2.020583e-01
##             HR_NAMEColorado River               HR_NAMENorth Coast
##                  -5.104965e-01                      2.026713e-01
##             HR_NAMENorth Lahontan          HR_NAMESacramento River
##                   6.182859e-01                     -9.136959e-03
##           HR_NAMESan Francisco Bay         HR_NAMESan Joaquin River
##                   1.986777e-02                     -4.189042e-01
##              HR_NAMESouth Coast               HR_NAMESouth Lahontan
##                  -3.326501e-01                     -2.413519e-01
##              HR_NAMETulare Lake
##                  -7.503123e-01
##
## Intercepts:
##                 At-Risk|Failing             Failing|Not At-Risk
##                      -1.3379340                      -0.6028577
## Not At-Risk|Potentially At-Risk
##                       1.7488469
##
## Residual Deviance: 5920.078
## AIC: 5956.078
```

```
CCR_anymissing <- glm(Missing_ccr_any ~ carceral_sys + population_served_count + watersource + purchase
summary(CCR_anymissing)
```

```
##
## Call:
## glm(formula = Missing_ccr_any ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                          Estimate Std. Error z value
## (Intercept)                            -9.362e-01  1.057e-01  -8.856
## carceral_syscarceral_serving_water_system -2.199e-01  2.456e-01  -0.895
## carceral_syscarceral_water_system       -1.494e-01  3.210e-01  -0.465
## population_served_count                -6.100e-06  1.860e-06  -3.279
## watersourceSW                          -3.789e-01  1.217e-01  -3.112
## purchasedPurchased                     -2.880e-01  1.550e-01  -1.858
## pws_type_codeNTNCWS                    -1.998e-01  1.056e-01  -1.893
## HR_NAMEColorado River                   1.047e+00  2.147e-01   4.878
## HR_NAMENorth Coast                      2.113e+00  1.587e-01  13.314
## HR_NAMENorth Lahontan                   3.006e+00  3.933e-01   7.642
## HR_NAMESacramento River                 2.289e+00  1.495e-01  15.306
## HR_NAMESan Francisco Bay                1.845e+00  1.828e-01  10.094
## HR_NAMESan Joaquin River                6.334e-01  1.360e-01   4.656
## HR_NAMESouth Coast                      6.816e-01  1.570e-01   4.342
## HR_NAMESouth Lahontan                  -3.665e-01  1.961e-01  -1.869
## HR_NAMETulare Lake                     -1.153e+00  1.869e-01  -6.168
##                                        Pr(>|z|)
## (Intercept)                             < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.37063
```

```
## carceral_syscarceral_water_system          0.64160
## population_served_count                    0.00104 **
## watersourceSW                              0.00186 **
## purchasedPurchased                         0.06320 .
## pws_type_codeNTNCWS                        0.05841 .
## HR_NAMEColorado River                     1.07e-06 ***
## HR_NAMENorth Coast                         < 2e-16 ***
## HR_NAMENorth Lahontan                     2.14e-14 ***
## HR_NAMESacramento River                    < 2e-16 ***
## HR_NAMESan Francisco Bay                   < 2e-16 ***
## HR_NAMESan Joaquin River                  3.23e-06 ***
## HR_NAMESouth Coast                        1.41e-05 ***
## HR_NAMESouth Lahontan                      0.06163 .
## HR_NAMETulare Lake                        6.94e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4689.7  on 3441  degrees of freedom
## Residual deviance: 3768.6  on 3426  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 3800.6
##
## Number of Fisher Scoring iterations: 6
```

```
CCR_missingtwoplus <- glm(Missing_ccr_twoplus ~ carceral_sys + population_served_count + watersource + p
summary(CCR_missingtwoplus)
```

```
##
## Call:
## glm(formula = Missing_ccr_twoplus ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -1.591e+00  1.298e-01 -12.252
## carceral_syscarceral_serving_water_system -7.253e-01  3.895e-01  -1.862
## carceral_syscarceral_water_system         -1.374e+00  4.704e-01  -2.921
## population_served_count                   -3.413e-05  6.140e-06  -5.559
## watersourceSW                             -2.146e-01  1.306e-01  -1.643
## purchasedPurchased                        -3.925e-01  1.930e-01  -2.034
## pws_type_codeNTNCWS                       -2.426e-01  1.142e-01  -2.125
## HR_NAMEColorado River                      9.759e-01  2.464e-01   3.961
## HR_NAMENorth Coast                         1.833e+00  1.669e-01  10.979
## HR_NAMENorth Lahontan                      3.050e+00  3.329e-01   9.161
## HR_NAMESacramento River                    2.344e+00  1.610e-01  14.560
## HR_NAMESan Francisco Bay                   1.849e+00  2.007e-01   9.210
## HR_NAMESan Joaquin River                   8.692e-01  1.598e-01   5.439
## HR_NAMESouth Coast                         1.926e-01  2.108e-01   0.914
## HR_NAMESouth Lahontan                     -1.238e+00  3.349e-01  -3.696
## HR_NAMETulare Lake                        -1.692e+00  3.003e-01  -5.633
##                                           Pr(>|z|)
## (Intercept)                                < 2e-16 ***
```

```
## carceral_syscarceral_serving_water_system 0.062555 .
## carceral_syscarceral_water_system         0.003490 **
## population_served_count                   2.71e-08 ***
## watersourceSW                             0.100366
## purchasedPurchased                        0.041931 *
## pws_type_codeNTNCWS                       0.033604 *
## HR_NAMEColorado River                     7.46e-05 ***
## HR_NAMENorth Coast                         < 2e-16 ***
## HR_NAMENorth Lahontan                      < 2e-16 ***
## HR_NAMESacramento River                    < 2e-16 ***
## HR_NAMESan Francisco Bay                   < 2e-16 ***
## HR_NAMESan Joaquin River                  5.37e-08 ***
## HR_NAMESouth Coast                        0.360747
## HR_NAMESouth Lahontan                     0.000219 ***
## HR_NAMETulare Lake                        1.77e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 4072.9  on 3441  degrees of freedom
## Residual deviance: 3096.7  on 3426  degrees of freedom
##   (817 observations deleted due to missingness)
## AIC: 3128.7
##
## Number of Fisher Scoring iterations: 8
```

```
Modeleddroughtimpact <- glm(SC5e_Drought_Impact ~ carceral_sys + population_served_count + watersource +
summary(Modeleddroughtimpact)
```

```
##
## Call:
## glm(formula = SC5e_Drought_Impact ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -3.711e+00  3.276e-01 -11.328
## carceral_syscarceral_serving_water_system  6.196e-01  6.174e-01   1.003
## carceral_syscarceral_water_system         -1.632e+01  1.381e+03  -0.012
## population_served_count                    3.808e-06  3.074e-05   0.124
## watersourceSW                              1.482e+00  2.077e-01   7.138
## purchasedPurchased                        -1.173e+00  4.436e-01  -2.644
## pws_type_codeNTNCWS                       -1.663e+01  4.160e+02  -0.040
## HR_NAMEColorado River                      8.628e-02  6.787e-01   0.127
## HR_NAMENorth Coast                         1.050e+00  3.957e-01   2.655
## HR_NAMENorth Lahontan                     -4.088e-01  1.071e+00  -0.382
## HR_NAMESacramento River                    1.035e+00  3.703e-01   2.796
## HR_NAMESan Francisco Bay                  -1.246e-01  6.143e-01  -0.203
## HR_NAMESan Joaquin River                   1.558e-01  4.268e-01   0.365
## HR_NAMESouth Coast                        -4.914e-01  6.119e-01  -0.803
## HR_NAMESouth Lahontan                      4.703e-01  5.062e-01   0.929
## HR_NAMETulare Lake                         1.179e+00  3.904e-01   3.020
##                                           Pr(>|z|)
```

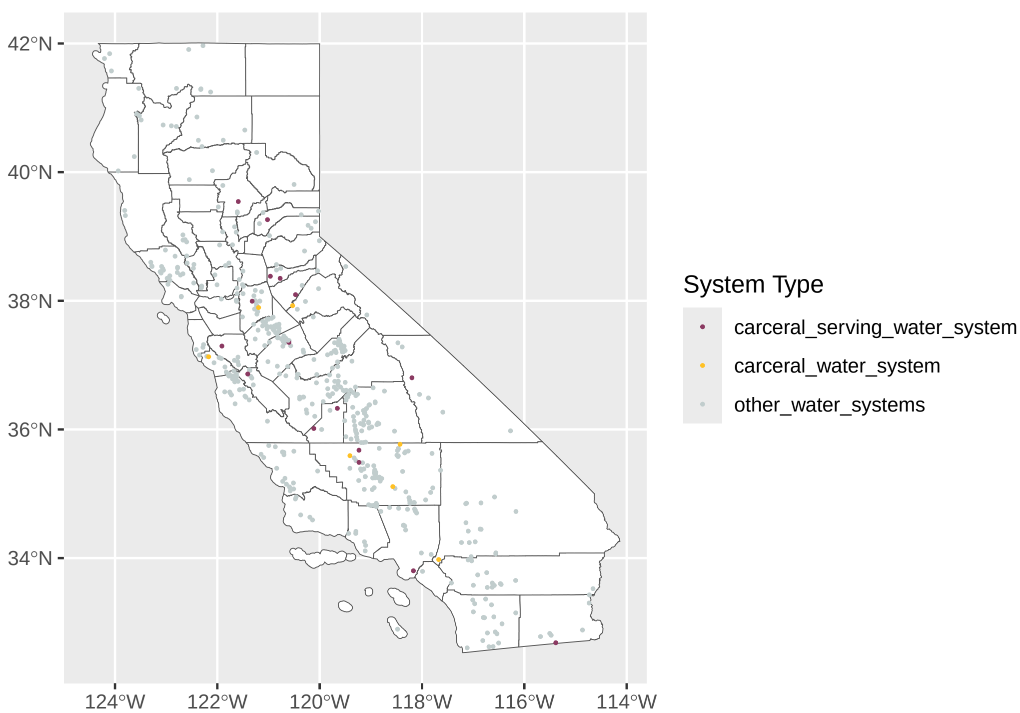
```
## (Intercept)                              < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.31562
## carceral_syscarceral_water_system          0.99057
## population_served_count                    0.90142
## watersourceSW                             9.48e-13 ***
## purchasedPurchased                         0.00819 **
## pws_type_codeNTNCWS                        0.96810
## HR_NAMEColorado River                      0.89885
## HR_NAMENorth Coast                         0.00793 **
## HR_NAMENorth Lahontan                      0.70278
## HR_NAMESacramento River                    0.00518 **
## HR_NAMESan Francisco Bay                   0.83931
## HR_NAMESan Joaquin River                   0.71498
## HR_NAMESouth Coast                         0.42197
## HR_NAMESouth Lahontan                      0.35284
## HR_NAMETulare Lake                         0.00252 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1059.99  on 2884  degrees of freedom
## Residual deviance:  895.03  on 2869  degrees of freedom
##   (1374 observations deleted due to missingness)
## AIC: 927.03
##
## Number of Fisher Scoring iterations: 18
```

```r
Distribution_disrupt <- glm(Distributiondis_any ~ carceral_sys + population_served_count + watersource +
summary(Distribution_disrupt)
```

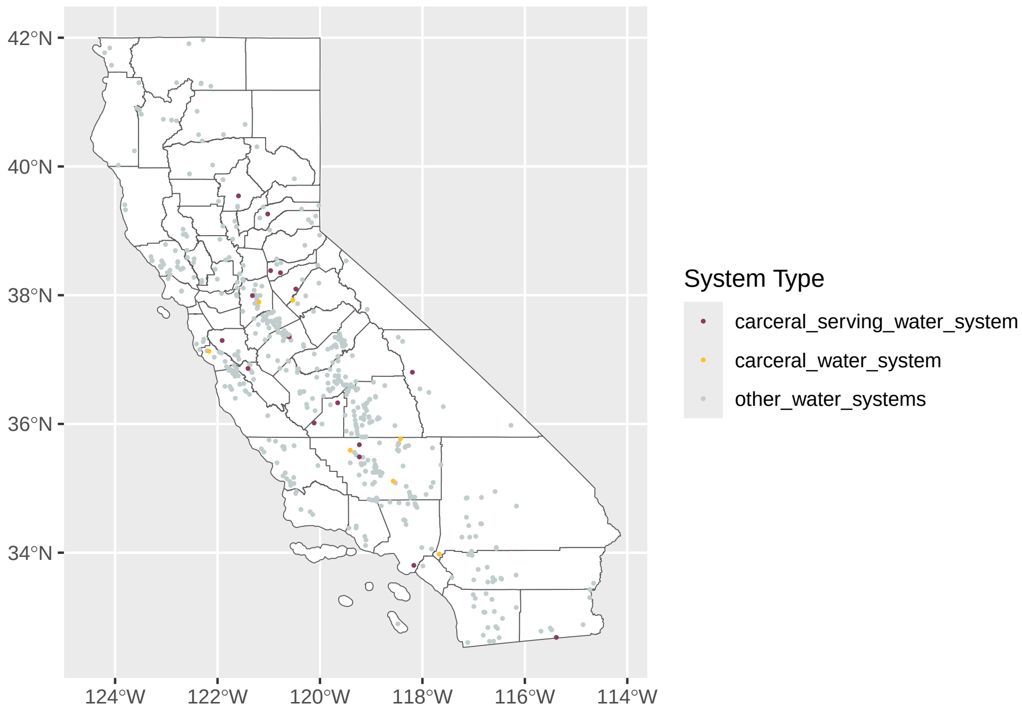
```
##
## Call:
## glm(formula = Distributiondis_any ~ carceral_sys + population_served_count +
##     watersource + purchased + pws_type_code + HR_NAME, family = binomial,
##     data = Data)
##
## Coefficients:
##                                             Estimate Std. Error z value
## (Intercept)                               -1.960e+00  1.582e-01 -12.391
## carceral_syscarceral_serving_water_system  4.522e-01  4.902e-01   0.922
## carceral_syscarceral_water_system         -1.856e+00  1.016e+00  -1.827
## population_served_count                    1.789e-05  9.631e-06   1.857
## watersourceSW                              5.269e-01  1.658e-01   3.179
## purchasedPurchased                         5.217e-02  2.315e-01   0.225
## pws_type_codeNTNCWS                       -1.763e+00  2.765e-01  -6.375
## HR_NAMEColorado River                     -1.051e-01  3.599e-01  -0.292
## HR_NAMENorth Coast                         8.014e-02  2.364e-01   0.339
## HR_NAMENorth Lahontan                     -4.134e-03  4.670e-01  -0.009
## HR_NAMESacramento River                    9.002e-02  2.115e-01   0.426
## HR_NAMESan Francisco Bay                   1.844e-01  2.874e-01   0.642
## HR_NAMESan Joaquin River                   3.116e-02  2.143e-01   0.145
## HR_NAMESouth Coast                        -1.431e-01  2.570e-01  -0.557
## HR_NAMESouth Lahontan                     -6.206e-01  3.207e-01  -1.935
## HR_NAMETulare Lake                        -6.229e-01  2.718e-01  -2.292
```

```
##                                              Pr(>|z|)
## (Intercept)                                   < 2e-16 ***
## carceral_syscarceral_serving_water_system  0.35628
## carceral_syscarceral_water_system          0.06764 .
## population_served_count                    0.06330 .
## watersourceSW                              0.00148 **
## purchasedPurchased                         0.82168
## pws_type_codeNTNCWS                       1.83e-10 ***
## HR_NAMEColorado River                      0.77033
## HR_NAMENorth Coast                         0.73459
## HR_NAMENorth Lahontan                      0.99294
## HR_NAMESacramento River                    0.67041
## HR_NAMESan Francisco Bay                   0.52118
## HR_NAMESan Joaquin River                   0.88440
## HR_NAMESouth Coast                         0.57760
## HR_NAMESouth Lahontan                      0.05302 .
## HR_NAMETulare Lake                         0.02193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1947.3  on 2847  degrees of freedom
## Residual deviance: 1824.2  on 2832  degrees of freedom
##   (1411 observations deleted due to missingness)
## AIC: 1856.2
##
## Number of Fisher Scoring iterations: 6
```

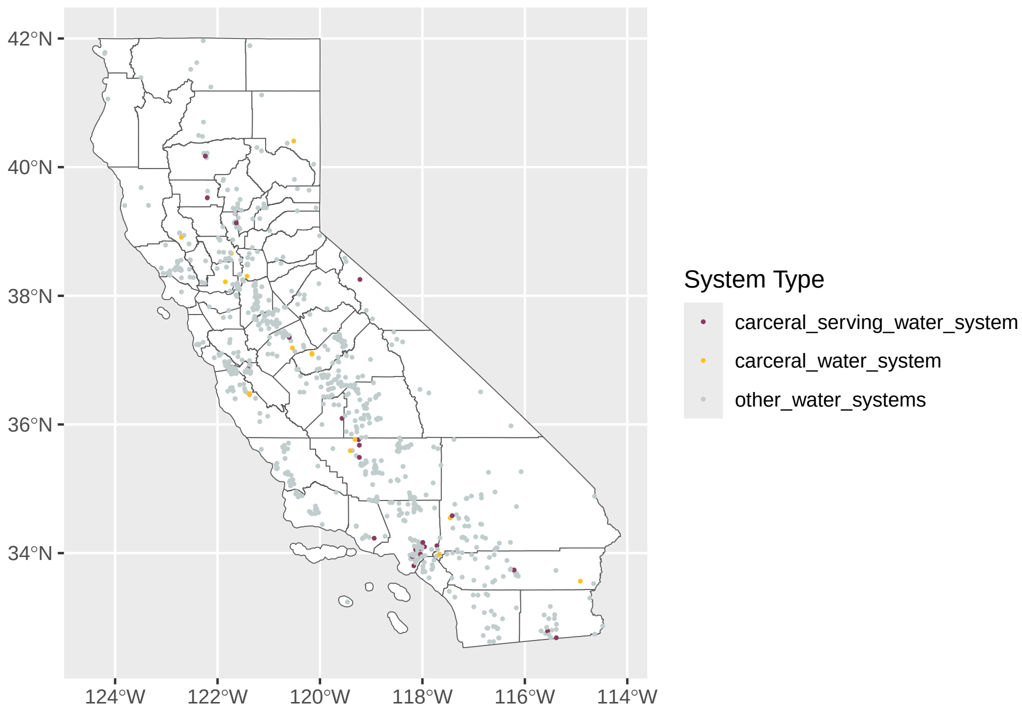
Now I want to see if there are any geographical patterns based on performance - In order MCL violations, monitoring violations, quality risk, accessibility risk, TMF risk, safer status



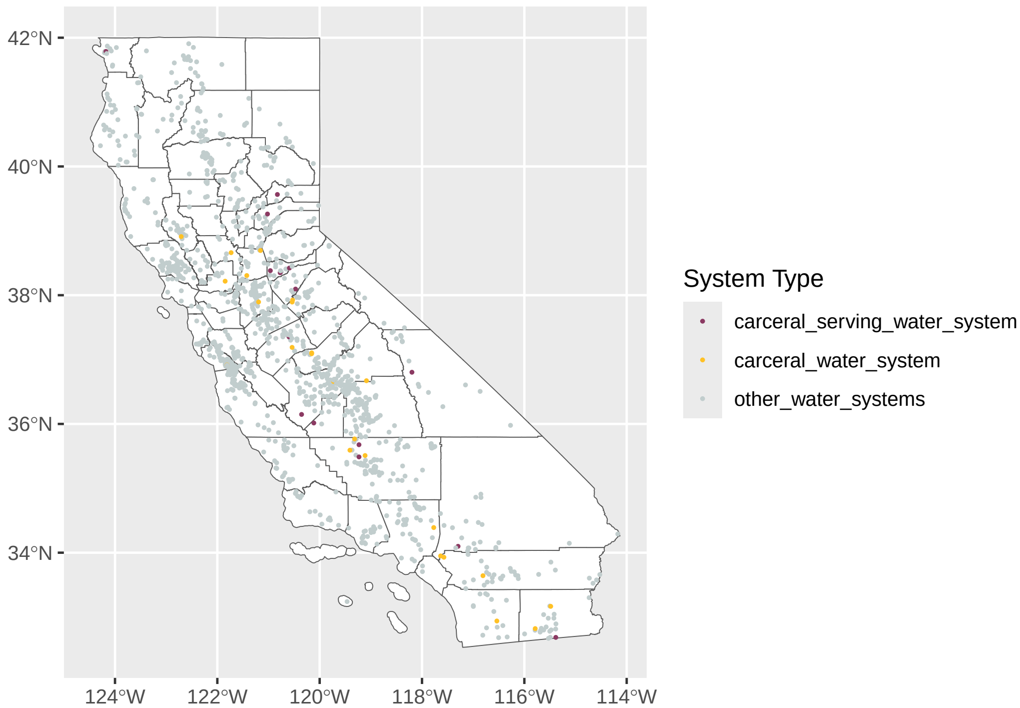
map by outcomes-1.pdf



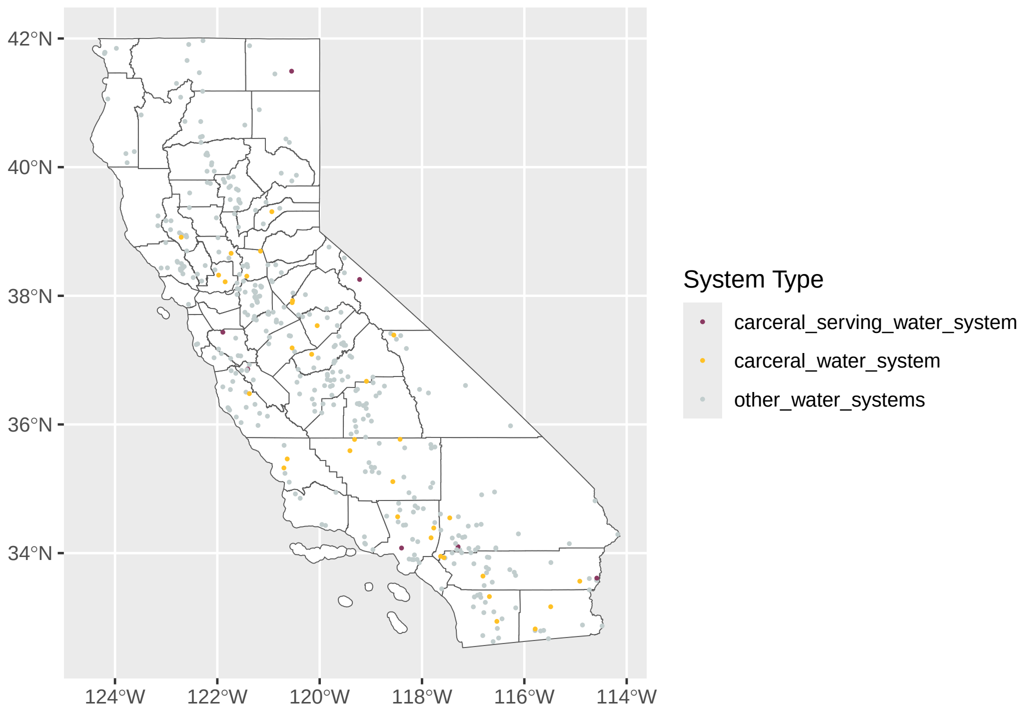
map by outcomes-2.pdf



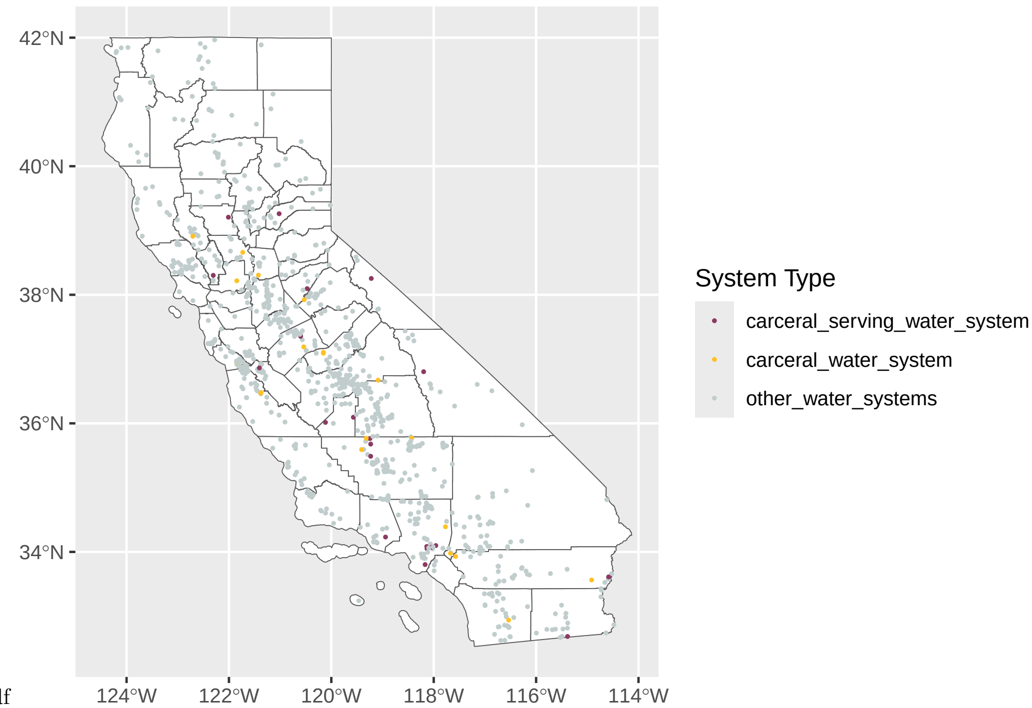
map by outcomes-3.pdf



map by outcomes-4.pdf



map by outcomes-5.pdf



map by outcomes-6.pdf